

mtejas619@gmail.com
(+91) 87673 83170
Indian Citizen

Tejas Mankeshwar

LinkedIn/tejasmanankeshwar
GitHub/TejasMankeshwar
Portfolio

EDUCATION

Savitribai Phule University

B.E in Computer Science, GPA: 9.2/10

Pune, MH

Sept 2022 – May 2026

- **Concentrations:** Artificial Intelligence & Machine Learning
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Objects & Software Design, Cloud Computing, Systems & Networks, Design & Analysis of Algorithms, Database Systems, Statistics, Advance Mathematics, Big Data Analytics, High Performance Computing, Natural Language Processing

WORK EXPERIENCE

ProjectPro

Data Science Intern

Bengaluru, KA

Oct 2024 – Jan 2025

- **Video Analyzer:** Designed and implemented a multimodal video analysis pipeline using OpenCV and MediaPipe to extract structured insights from text, audio, and visual streams.
- **Blog Generation Model:** Built an SEO-oriented blog generation system leveraging OpenAI and Gemini GPT APIs, integrating Wikipedia and ArXiv agents to ensure factual consistency and reliable knowledge grounding.
- **Image Generation Model (Picture Book):** Developed an AI-powered image generation pipeline using Stable Diffusion and PyTorch to create illustrated visual narratives for automated storytelling applications.
- **Article Classifier (Rule-Based):** Engineered a rule-based NLP article classification system that accurately categorized content without supervised ML models, reducing computational overhead while maintaining high precision.

MIT-WPU Institute of Artificial Intelligence

Artificial Intelligence Research Intern

Pune, MH

August 2024 – Dec 2025

- Implemented the Snail Homing and Mating Search (SHMS) algorithm in code, translating a mentor-proposed optimization concept into a complete, working computational framework.
- Extended SHMS to multi-objective optimization, benchmarking performance across ZDT test suites and analyzing Pareto-front convergence and diversity.
- Authored peer-reviewed chapters on *Normal Distribution* and *Weighted Combination Models* to Springer's 'The Encyclopedia of Engineering Optimization and Heuristics'

PROJECTS

AI-Driven Molecular Generation Pipeline (B.E. Capstone Project)

GitHub

Python, PyTorch, Reinforcement Learning, RDKit, SMILES

- Designed and developed an end-to-end AI pipeline that ingests molecules in standardized SMILES format (CSV) and autonomously generates novel drug-like molecules with optimized properties.
- Architecture comprises three parallel components: a **Generator Engine** (analyzes molecular structure and generates new candidates), a **Surrogate Model** (predicts molecular properties and assigns scores), and a **Reinforcement Learning module** (evaluates generated molecules and feeds reward signals back to the generator).
- **My Contribution:** Orchestrated, designed, and implemented all three ML/AI models from scratch. Currently serving as **Team Lead**, overseeing end-to-end project execution, task delegation, and technical direction across the team.

Trading Intel — AI-Powered Stock Analysis Platform

GitHub

Python, FastAPI, React, Scikit-learn, Random Forest, Pandas, yfinance, Recharts

- Developed a full-stack AI platform delivering deterministic stock price projections for Nifty 50 companies across multiple time horizons (1W, 1M, 3M) using a **Random Forest Regressor** ensemble.
- Achieved **80% prediction accuracy** during testing phase, validated through backtesting on 2 years of historical NSE data; engineered features include MACD, RSI, Bollinger Band Width, and EMA.
- Built a high-performance **FastAPI** backend with an in-memory caching mechanism for low-latency responses, and a responsive **React** dashboard visualizing actual vs. predicted price trends with custom confidence scoring.

VistaDrive — Scenic Route Generator

GitHub

Python, OSMnx, NetworkX, Streamlit, Folium, GeoPy, GeoPandas, Google Maps API

- Built a geospatial web application that generates optimized scenic driving loops of user-defined distances, prioritizing primary/secondary road networks extracted from **OpenStreetMap** via **OSMnx**.
- Engineered a graph-based pathfinding engine using **NetworkX** (Dijkstra's algorithm) with a geometric waypoint generation algorithm using spherical bearing-distance calculations to produce seamless 4-point loop routes.
- Integrated **Folium** for interactive spatial visualization; implemented a caching layer for road graph data to reduce API latency, with export support for KML files and Google Maps deep-linking.

Chess Simulator & AI Analysis Tool

GitHub

Python, Pygame, python-chess, Stockfish (UCI Protocol)

- Developed a fully functional desktop chess simulator with a responsive 2D GUI using **Pygame**, featuring intuitive drag-and-drop mechanics, board rotation, and a move-undo system.
- Integrated the **Stockfish UCI engine** for real-time best-move suggestions and positional evaluation; implemented a dynamic evaluation bar visualizing material/positional advantage (centipawns) with mate-threat detection.
- Handled asynchronous engine communication to prevent GUI lag, maintaining 60 FPS; automated asset pipeline via **urllib** to fetch and process public-domain piece assets from Wikimedia Commons.

PUBLICATIONS

- **Tejas Mankeshwar (2025).** “Normal Distribution.”: Book chapter authored for *The Encyclopedia of Engineering Optimization and Heuristics*, Springer. Covers mathematical foundations, classifications, and applications in statistics and AI/ML.
- **Tejas Mankeshwar (2025).** “Weighted-Sum Method.”: Book chapter authored for *The Encyclopedia of Engineering Optimization and Heuristics*, Springer. Discusses scalarization techniques for multi-objective optimization, Pareto fronts, and engineering applications.
- **Prajakta Gunjal, Tejas Mankeshwar, Girija Chiddarwar, Bhushan Anokar (2025).**: *High-Accuracy Classification of Pancreatic Cancer Subtypes Using Feature Selection and Optimized Machine Learning on High-Throughput Gene Expression Data*. Research paper proposing an optimized ML-based biomedical classification pipeline.
- **Tejas Mankeshwar, Girija Chiddarwar, Anuj Dengale, Anvita Kashikar, Esha Ratnaparkhi (2025).**: *Transforming Drug Discovery and Development: The Synergistic Role of AI, Machine Learning, Deep Learning, and Quantum Computing*. Review paper on AI/ML and quantum computing across the drug discovery pipeline.
- **Varad Marne, Yash Khetal, Tejas Mankeshwar, Soham Miniyar (2023).**: *Review of Dynamic Wireless Power Transfer Systems for Electric Vehicles and Its Application in India*. Review paper analyzing DWPT technologies and deployment challenges in India.

ACHIEVEMENTS & HONORS

- **Winner — PES Modern College Hackathon:** Built an AI-enabled social NGO emergency interconnection platform using Google Earth and LLM-based services to enable real-time coordination during crises.
- **2nd Place — MIT-ADT University Ideathon:** Developed *Legal-Saathi*, an AI-powered legal aid platform to help citizens understand Indian law and judiciary processes, featuring document upload and simplified legal explanations for non-technical users.
- **3rd Place — IMCC Hackathon:** Created *NeXstep*, a demand forecasting and smart pricing system for retail and wholesale inventory optimization using demand probability modeling and competitor price analysis.
- **Multiple Hackathon Winner:** Secured **8+ hackathon wins** across national-level competitions, recognized for innovation in AI-driven platforms, optimization, and real-world problem solving.

SKILLS

Programming Languages: Python, C++, C, Java, JavaScript, R, MATLAB

AI / Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Reinforcement Learning, Feature Selection, Model Optimization, Statistical Modeling

Frameworks & Libraries: PyTorch, Hugging Face Transformers, OpenCV, Flask, Streamlit

Web & Backend Technologies: FastAPI, React, JavaScript, HTML, CSS, Bootstrap

Databases & Search: MySQL, MongoDB, Elasticsearch

Cloud & DevOps: AWS (EC2, Lambda), Docker, Linux, Git, Google Colab

Optimization & Algorithms: Multi-Objective Optimization, Heuristic Algorithms, MOSHMS, ZDT, DTLZ Benchmarks, Pareto Optimization

Tools & Platforms: GitHub, VS Code, Jupyter Notebook, LaTeX